

线性回归

1.1 基本模型 $h_w(x) = w^T x + b$.

为简化模型, $x \leftarrow \begin{bmatrix} x \\ 1 \end{bmatrix}$, $w \leftarrow \begin{bmatrix} w \\ b \end{bmatrix}$.

则 $h_w(x) = w^T x$

1.2. 最小二乘法:

目标: 最小化平方误差: $J(w) = \frac{1}{2} \sum_{i=1}^n (h_w(x_i) - y_i)^2$.

以矩阵表示:

$$X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix} \in \mathbb{R}^{n \times (d+1)}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

—— 一个向量, n 维: y_n

$$J(w) = \frac{1}{2} (Xw - y)^T (Xw - y).$$

1.3 推导: $J(w) = \frac{1}{2} (w^T X^T - y^T X) (Xw - y)$

$$= \frac{1}{2} (w^T X^T X w - 2 w^T X^T y + y^T y)$$

注: 常用公式:

梯度. 标量 $f(x)$ 对列向量 x 的导数

$$\frac{\partial f}{\partial x} = \nabla_x f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

Jacobian 矩阵: 向量 $f(x)$ 对列向量 x 的导数.

若 $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, 则 $J \in \mathbb{R}^{m \times n}$, $J_{ij} = \frac{\partial f_i}{\partial x_j}$

$$\frac{\partial \vec{f}}{\partial \vec{x}} = \vec{J} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

Hessian矩阵: 标量 $f(\vec{x})$ 对列向量 $\vec{x}$ 的二阶导数.

$$H(f) = \frac{\partial^2 f}{\partial x \partial x^T} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{bmatrix}$$

线性形式: $\frac{\partial (a^T x)}{\partial x} = \frac{\partial (x^T a)}{\partial x} = a.$

二次型 $\frac{\partial (x^T A x)}{\partial x} = (A + A^T) x$. 当 $A = A^T$, $\frac{\partial x^T A x}{\partial x} = 2Ax$.

2范平方: $\frac{\partial \|x\|_2^2}{\partial x} = \frac{\partial x^T x}{\partial x} = 2x.$

复合函数: 令 $z = f(\vec{y})$, $\vec{y} = g(\vec{x})$

$$\frac{\partial z}{\partial \vec{x}} = \frac{\partial \vec{y}}{\partial \vec{x}}, \quad \frac{\partial z}{\partial \vec{y}} = \left(\frac{\partial \vec{y}}{\partial \vec{x}} \right)^T \frac{\partial z}{\partial \vec{y}} = \vec{J}_y^T \nabla_y z.$$

线性变换: $\frac{\partial (Ax)}{\partial x} = A.$

迹的线性: $\frac{\partial \text{tr}(AX)}{\partial X} = A^T$

迹的二次型 $\frac{\partial \text{tr}(X^T A X)}{\partial X} = (A + A^T) X$ 若 A 对称, 则为 $2AX$.

行列式对数: $\frac{\partial \log |X|}{\partial X} = (X^{-1})^T$

Frobenius范平方: $\frac{\partial \|X\|_F^2}{\partial X} = \frac{\partial \text{tr}(X^T X)}{\partial X} = 2X$

续: 1.3.

$$\text{求导, 得梯度: } \frac{\partial J}{\partial w} = \frac{1}{2} \frac{\partial (w^T X^T X w)}{\partial w} - \frac{\partial (w^T X^T \bar{y})}{\partial w}$$

$$= X^T X w - X^T \bar{y}$$

$$\text{令 } \frac{\partial J}{\partial w} = 0, \quad X^T X w - X^T \bar{y} = 0.$$

$$\Rightarrow X^T X w = X^T \bar{y}.$$

$$\Rightarrow w^* = (X^T X)^{-1} X^T \bar{y}.$$

此为 $X^T X$ 可逆时的解析解.

二、带特征映射的线性回归.

2.1 当数据非线性可分, 引入特征映射 $\varphi: \mathbb{R}^d \rightarrow \mathbb{R}^m$ ($m \geq d$).

$$\varphi(\vec{x}) = [\varphi_1(\vec{x}), \dots, \varphi_m(\vec{x})]^T.$$

$$\text{多项式映射: } \varphi(\vec{x}) = [1, x_1, x_2, x_1^2, x_1 x_2, x_2^2, \dots]^T.$$

$$\text{径向基函数: } \varphi_j(\vec{x}) = \exp\left(-\frac{\|\vec{x} - \vec{g}_j\|^2}{2\sigma^2}\right)$$

2.2 映射后: $h_w(\vec{x}) = w^T \varphi(\vec{x})$

$$\Phi = \begin{bmatrix} \varphi(\vec{x}_1)^T \\ \vdots \\ \varphi(\vec{x}_n)^T \end{bmatrix} \in \mathbb{R}^{n \times m}$$

$$\text{解析解: } w^* = (\Phi^T \Phi)^{-1} \Phi^T y.$$

三、岭回归与正则化.

3.1 当 $m > n$ (特征维数 > 样本数) 时, $\Phi^T \Phi$ 不可逆或病态, 需要正则化.

3.2 岭回归 (Ridge Regression)

$$\text{目标函数: } J(w) = \frac{1}{2} \sum_{i=1}^n (h_w(\vec{x}_i) - y_i)^2 + \frac{\lambda}{2} \|w\|_2^2$$

$$\begin{aligned}
&= \frac{1}{2} \|\Phi w - y\|^2 + \frac{\lambda}{2} \|w\|^2 \\
&= \frac{1}{2} (\Phi w - y)^T (\Phi w - y) + \frac{1}{2} w^T w \\
&= \frac{1}{2} w^T \Phi^T \Phi w - w^T \Phi^T y + \frac{1}{2} y^T y + \frac{1}{2} w^T w.
\end{aligned}$$

$$\text{令 } \frac{\partial J}{\partial w} = \Phi^T \Phi w - \Phi^T y + \lambda w = 0.$$

$$(\Phi^T \Phi + \lambda I) w = \Phi^T y$$

$$w^* = (\Phi^T \Phi + \lambda I)^{-1} \Phi^T y$$

其中 $\lambda > 0$, 保证 $(\Phi^T \Phi + \lambda I)$ 可逆.

可以约束 $\|w\|$ 不过大, 防止过拟合.

四. 贝叶斯视角的正则化

4.1 贝叶斯定理: $p(w|D) = \frac{p(D|w)p(w)}{p(D)} \propto p(D|w)p(w).$

4.2 岭回归的贝叶斯解释.

Assumption 1: $y_i = w^T \phi(x_i) + \epsilon_i, \epsilon_i \sim \mathcal{N}(0, \sigma^2).$

似然函数: $p(y|x, w) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - w^T \phi(x_i))^2}{2\sigma^2}\right)$

Assumption 2: $w \sim \mathcal{N}(0, \tau^2 I)$ 先验 $p(w) = \frac{1}{(2\pi\tau^2)^{\frac{p}{2}}} \exp\left(-\frac{1}{2\tau^2} \|w\|^2\right)$

后验分布: $p(w|x, y) \propto p(y|x, w)p(w)$

$$\log p(w|x, y) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - w^T \phi(x_i))^2 - \frac{1}{2\tau^2} \|w\|_2^2 + C$$

最大化后验 (MAP): 向量化

$$w_{\text{MAP}} = \arg \max_w \left[-\frac{1}{2\sigma^2} \|\Phi w - y\|^2 - \frac{1}{2\tau^2} \|w\|^2 \right]$$

$$\Leftrightarrow w_{\text{map}} = \arg \min_w [\|\Phi w - y\|^2 + \frac{\sigma^2}{\epsilon^2} \|w\|^2]$$

似然: $p(y|X, w)$ 已知 X, w , 观测 y 的概率,

$y|X, w \sim N(\Phi w, \sigma^2 I)$ (高斯噪声).

先验: $p(w)$ 对参数 w 的分布假设 $w \sim N(0, \sigma^2 I)$ 0均值高斯先验,

后验: $p(w|X, y)$. 已知输入 X , 输出 y , w 的概率分布.

这就是MAP要最大化的.

后验 $\propto$ 先验 $\times$ 似然.

最小二乘可以看作一个求解 w 的先验的过程.

岭回归就是在高斯先验下的最小二乘

故此处我们最大化后验的概率

也就是最小化后验的概率的相反数.



对比岭回归: $\lambda = \frac{\sigma^2}{\epsilon^2}$.

λ 大, 则 σ^2 (先验) 小, 权重集中在 0 附近
 λ 小, σ^2 大, 权重先验分布较平坦

} 正则强弱

五、超参数确定:

5.1 交叉验证 λ .

K折交叉验证: 1. 将数据分为 K 份

2. 用 $K-1$ 份训练, 1 份验证.

3. 计算平均验证误差.

4. 选择 $\lambda^* = \arg \min_{\lambda} (V\text{-Error}(\lambda))$

5.2 贝叶斯法确定 λ .

$$p(y|X, \lambda) = \int p(y|X, w) p(w|\lambda) dw.$$

对于高斯先验与似然,有解析解:

$$\log p(y|X, D) = -\frac{1}{2} n \log 2\pi - \log |C| - \frac{1}{2} y^T C^{-1} y$$

$$C = \sigma^2 I + \Phi \Lambda^{-1} \Phi^T, \Lambda = \frac{1}{\sigma^2} I.$$

六. 梯度下降法.

6.1 批梯度下降.

$$W_{k+1} = W_k - \alpha \frac{\partial J(W_k)}{\partial W} \quad (\text{几乎所有监督学习都适用}).$$

核心: $W \leftarrow W - \alpha$ (与梯度相关向量),

对于线性回归, $J(W) = \frac{1}{2} \|\Phi W - y\|_2^2$

$$\frac{\partial J}{\partial W} = \Phi^T \Phi W - \Phi^T y$$

故 $W_{t+1} = W_t - \alpha (\Phi^T \Phi W_t - \Phi^T y)$,

$$\lim_{t \rightarrow \infty} W_t = (\Phi^T \Phi)^{-1} \Phi^T y.$$

① 迭代 $\rightarrow$ 线性收敛: 证明 $W_{k+1} - W^* = A(W_k - W^*)$ 即误差为线性迭代

② A 为幂零矩阵/对称阵等, 误差会在有限步归零.

③ 选择合适的步长 α . 让 $A^m = 0$ 对某个有限 m 成立.

④ 结论: 误差在 m 步后收敛到 0

对本例 令 $e_k = W_k - W^*$ $\swarrow$ 最优解

$$\begin{aligned} W_{k+1} &= W_k - \alpha \Phi^T (\Phi W_k - y) \\ &= W_k - \alpha \Phi^T \Phi W_k + \alpha \Phi^T y. \end{aligned}$$

$$= (\mathbf{I} - \alpha \Phi^T \Phi) \mathbf{w}_k + \alpha \Phi^T \mathbf{y}.$$

$$\mathbf{w}_{k+1} - \mathbf{w}^* = (\mathbf{I} - \alpha \Phi^T \Phi) \mathbf{w}_k + \alpha \Phi^T \mathbf{y} - \mathbf{w}^*.$$

$$\because \Phi^T \mathbf{y} = \Phi^T \Phi \mathbf{w}^*.$$

$$\text{故 } \mathbf{w}_{k+1} - \mathbf{w}^* = (\mathbf{I} - \alpha \Phi^T \Phi) \mathbf{w}_k + (\alpha \Phi^T \Phi - \mathbf{I}) \mathbf{w}^*.$$

$$\text{故 } \mathbf{e}_{k+1} = \mathbf{A} \mathbf{e}_k. \text{ 此处 } \mathbf{I} - \alpha \Phi^T \Phi = \mathbf{A}.$$

$$\mathbf{e}_k = \mathbf{A}^k \mathbf{e}_0.$$

$\Phi^T \Phi$ 为对称半正定, 若 $\Phi^T \Phi$ 满秩, 则 $\Phi^T \Phi = \mathbf{Q} \Lambda \mathbf{Q}^T$

$$\text{于是 } \mathbf{A} = \mathbf{Q} (\mathbf{I} - \alpha \Lambda) \mathbf{Q}^T$$

若对于有限 k 有 $\mathbf{A}^k = \mathbf{0}$, 则 $\mathbf{I} - \alpha \Lambda$ 为 幂零矩阵, 即对角元均为 0.

即 $1 - \alpha \lambda_i = 0$, 也就是说 $\Phi^T \Phi$ 只有一个非零特征值.

$$\text{rank}(\Phi) = 1$$

若 Φ 为一列, 则 $\alpha = \frac{1}{\lambda_1}$, $\mathbf{A} = \mathbf{0}$ $\mathbf{e}_1 = \mathbf{A} \mathbf{e}_0 = \mathbf{0}$, 一步收敛

若 Φ 满秩, $\text{rank}(\Phi) = d > 1$.

无法使 $1 - \alpha \lambda_i = 0$ 对 $\forall i$ 均成立.

但误差会指数下降, 渐近收敛到 0.

6.2. 随机梯度下降

不用全部样本, 只随机一个样本 i , 用它的梯度近似总体梯度.

$$\text{定义 } J_i(\mathbf{w}) = \frac{1}{2} (\phi_i^T \mathbf{w} - y_i)^2.$$

$$\frac{\partial J_i(\mathbf{w})}{\partial \mathbf{w}} = \phi_i (\phi_i^T \mathbf{w} - y_i).$$

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \alpha \cdot \frac{\partial J_i(\mathbf{w})}{\partial \mathbf{w}}$$

为什么? $E[\frac{\partial J_i(w)}{\partial w}] = \frac{1}{N} \sum_{i=1}^N \frac{\partial J_i(w)}{\partial w} = \nabla J(w)$ 无偏估计.
简单随机抽样.

每次迭代仅用单样本,

有噪声, 但更快, 可在线学习 (学完即弃, 不存历史数据).

需要递减学习率 $\alpha_t = \frac{\alpha_0}{1+kt}$

此处见随机过程 Robbins-Monro 条件:

1. $\sum_{t=1}^{\infty} \alpha_t = \infty$, 保证全局最优化

2. $\sum_{t=1}^{\infty} \alpha_t^2 < \infty$, 噪声被压制, 最终收敛.

$\alpha_t \approx \frac{c}{t}$. $\sum_{t=1}^{\infty} \frac{1}{t} \rightarrow \infty$ 可最优 $\sum_{t=1}^{\infty} \frac{1}{t^2} = \frac{\pi^2}{6} < \infty$ 噪声可控.

